

Lecture 4: Concentration Bounds for Matrices

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Recall sub-Gaussian random variables

Definition 1 (Sub-Gaussian random variable)

A random variable X is **sub-Gaussian** if there exist constants $C, v > 0$ such that for all $t > 0$:

$$\mathbb{P}(|X| > t) \leq Ce^{-vt^2}.$$

Intuition

Sub-Gaussian variables have **Gaussian-like tails**: the probability of large deviations decays **at least as fast as for a Gaussian**.

Definition 2 (Sub-Gaussian norm)

The sub-Gaussian norm of X is given by

$$\|X\|_{\psi_2} = \inf\{s > 0 \mid \mathbb{E}e^{(X/s)^2} \leq 2\}.$$

(Exercise: Verify that $X \sim \mathcal{N}(0, \sigma) \Rightarrow \|X\|_{\psi_2} = \sqrt{8/3}\sigma$)

Motivation

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, one is often interested in getting a handle on various statistics such as the eigenvalues or the **operator norm** as a function of the properties of the entries of $\mathbf{A}$

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$$\|\mathbf{A}\|_{\text{op}} = \sup_{\mathbf{x}: \|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|$$

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We start from the observation that the tail event $\{\|\mathbf{A}\|_{\text{op}} > t\}$ can be expressed as a union of many simpler events:

$$\Pr(\|\mathbf{A}\|_{\text{op}} > t) \leq \Pr\left(\bigcup_{\mathbf{x} \in \mathcal{S}^{n-1}} \|\mathbf{A}\mathbf{x}\| > t\right),$$

where $\mathcal{S}^{n-1} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$ is the unit sphere in $\mathbb{R}^n$

Bounding $\Pr(\|\mathbf{Ax}\| > t)$

the first step to bound the RHS of

$$\Pr(\|\mathbf{A}\|_{\text{op}} > t) \leq \Pr(\cup_{\mathbf{x} \in \mathcal{S}^{n-1}} \|\mathbf{Ax}\| > t)$$

is using **standard concentration inequalities**, such as the following one:

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Lemma 1

Suppose that the entries A_{ij} of $\mathbf{A}$ are independent, with mean zero, and uniformly bounded in magnitude by 1. Let $\mathbf{x}$ be a unit vector in $\mathbb{R}^n$. Then for sufficiently large constant t , one has

$$\Pr(\|\mathbf{Ax}\| \geq t\sqrt{n}) \leq C \exp(-ct^2n),$$

for some constants $C, c > 0$

Lemma Proof: Step 1 – Understanding $\mathbf{Ax}$

- $\mathbf{x}$ is a **unit vector**: $\sum_{j=1}^n |x_j|^2 = 1$.
- Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be the rows of $\mathbf{A}$. Then the column vector $\mathbf{Ax}$ has entries:

$$(\mathbf{Ax})_i = \mathbf{a}_i \cdot \mathbf{x}, \quad i = 1, \dots, n.$$

- Writing $\mathbf{x} = (x_1, \dots, x_n)^\top$, we have

$$\mathbf{a}_i \cdot \mathbf{x} = \sum_{j=1}^n A_{ij} x_j.$$

- By **standard concentration results**¹, each inner product $\mathbf{a}_i \cdot \mathbf{x}$ is sub-Gaussian.

¹Hoeffding's Lemma: if $X_1, \dots, X_n$ are independent, bounded $X_i \in [a_i, b_i]$, and let $S_n := X_1 + \dots + X_n$. Then $\Pr(|S_n - \mathbb{E} S_n| \geq \lambda \sigma) \leq C e^{-c \lambda^2}$, with $\sigma^2 = \sum_i (b_i - a_i)^2$.

Lemma Proof: Step 2 – Sub-Gaussian Tail and Markov Argument

Sub-Gaussian tail bound (for each i)

$$\Pr(|\mathbf{a}_i \cdot \mathbf{x}| \geq \lambda) \leq Ce^{-c\lambda^2}, \quad \mathbb{E}[e^{c|\mathbf{a}_i \cdot \mathbf{x}|^2}] \leq C.$$

- Multiplying $\mathbb{E}[e^{c|\mathbf{a}_i \cdot \mathbf{x}|^2}] \leq C$ for all i (and using independence of the rows $\mathbf{a}_i$) gives

$$\mathbb{E}[e^{c\|\mathbf{Ax}\|^2}] = \mathbb{E}[e^{c\sum_i |\mathbf{a}_i \cdot \mathbf{x}|^2}] \leq C^n.$$

- Apply **Markov's inequality** to the nonnegative variable $Z = e^{c\|\mathbf{Ax}\|^2}$:

$$\Pr(\|\mathbf{Ax}\| \geq t\sqrt{n}) = \Pr(e^{c\|\mathbf{Ax}\|^2} \geq e^{ct^2n}) \leq \frac{\mathbb{E}[e^{c\|\mathbf{Ax}\|^2}]}{e^{ct^2n}} \leq C^n e^{-ct^2n}.$$

□

$\Pr(\|\mathbf{Ax}\| > t)$ is fine... but what about $\Pr\left(\bigcup_{\mathbf{x} \in \mathcal{S}^{n-1}} \|\mathbf{Ax}\| > t\right)$?

What we know so far

For each fixed unit vector $\mathbf{x}$, we have

$$\|\mathbf{Ax}\| = O(\sqrt{n}) \quad \text{with overwhelming probability.}$$

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- It is tempting to look back at

$$\Pr(\|\mathbf{A}\|_{\text{op}} > t) \leq \Pr\left(\bigcup_{\mathbf{x} \in \mathcal{S}^{n-1}} \{\|\mathbf{Ax}\| > t\}\right),$$

and conclude that $\|\mathbf{A}\|_{\text{op}} = O(\sqrt{n})$ also, by applying a **union bound** over $\mathcal{S}^{n-1}$.

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and conclude that $\|\mathbf{A}\|_{\text{op}} = O(\sqrt{n})$ also, by applying a **union bound** over $\mathcal{S}^{n-1}$.

- However, the sphere $\mathcal{S}^{n-1}$ is **uncountable** – there are infinitely many directions $\mathbf{x}$.

The core issue

Even if each event occurs with exponentially small probability, the union could well be everything!

Motivation: Introducing the ε -net idea

From uncountable to countable

We can restrict our attention to a **countable dense subset** of $\mathcal{S}^{n-1}$, because the map $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|$ is continuous.

- This makes the union countable – but still infinite, so the union bound remains useless.
- Luckily, $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|$ is not only continuous, it's **Lipschitz continuous** with constant $\|\mathbf{A}\|_{\text{op}}$:

$$|\|\mathbf{A}\mathbf{x}\| - \|\mathbf{A}\mathbf{y}\|| \leq \|\mathbf{A}\|_{\text{op}} \|\mathbf{x} - \mathbf{y}\|.$$

- This strong form of continuity means that nearby points on the sphere behave almost the same.

Key idea

We can therefore approximate the whole sphere $\mathcal{S}^{n-1}$ by a **finite dense subset**: an ε -net.

Definition of ϵ -net

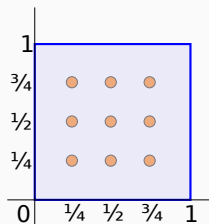
Definition 3 (ϵ -net)

Consider (T, d) as a metric space and $K \subset T$ a subset.

Let $\epsilon > 0$. A subset $K_\epsilon \subseteq K$ is an ϵ -net of K if **every point in K is within a distance ϵ of some point of K_ϵ** , i.e.

$$\forall \mathbf{x} \in K, \exists \mathbf{x}_0 \in K_\epsilon : d(\mathbf{x}, \mathbf{x}_0) \leq \epsilon.$$

An example of an $\frac{1}{2\sqrt{2}}$ -net (represented by the orange dots) in the **Euclidean distance** for the unit cube:



Covering number of the Euclidean sphere

Definition 4

The **smallest possible cardinality** of an ϵ -net of K is called the **covering number** of K and is denoted by $N(\epsilon, K)$.

Lemma 2 (Upper bound on $N(\epsilon, \mathcal{S}^{n-1})$)

The covering number of the n -dimensional unit Euclidean sphere

$$\mathcal{S}^{n-1} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_2 = 1\}$$

satisfies the following upper bound for any $\epsilon > 0$:

$$N(\epsilon, \mathcal{S}^{n-1}) \leq \left(\frac{2}{\epsilon} + 1\right)^n$$

Lemma Proof

- Consider a ϵ -net $\mathcal{N}_\epsilon$ of $\mathcal{S}^{n-1}$ such that, if we consider balls of radius $R_0 = \epsilon/2$ centered around each point of $\mathcal{N}_\epsilon$, these balls are **disjointed**.
- **These balls are contained in the ball of radius $R_1 = 1 + \frac{\epsilon}{2}$ centered at the origin.** Denote by $V(r)$ the volume of a ball of radius $r > 0$. Then, we have

$$|\mathcal{N}_\epsilon| \cdot V(R_0) \leq V(R_1)$$

- Recall that volume of the n -dimensional Euclidean ball of radius R is

$$V(R) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^n$$

- Therefore, the ratio of the volume of the small balls of radius R_0 over the large ball of radius R_1 is

$$|\mathcal{N}_\epsilon| \leq \frac{V(R_1)}{V(R_0)} = \left(\frac{1 + \frac{\epsilon}{2}}{\frac{\epsilon}{2}} \right)^n = \left(1 + \frac{2}{\epsilon} \right)^n$$

□

Computing the spectral/operator norm on a net(I)

We mentioned earlier that ϵ -nets give us a way to deal with the problem of an uncountable number of events when taking a union bound over the sphere. We want to **exploit Lipschitz continuity of the map $\mathbf{x} \mapsto \|\mathbf{Ax}\|_2$ to control what happens between the points in the ϵ -net.**

Idea

If the map $\mathbf{x} \mapsto \|\mathbf{Ax}\|_2$ is Lipschitz on $\mathcal{S}^{n-1}$ with constant $\|\mathbf{A}\|_{\text{op}}$, then knowing $\|\mathbf{Ax}\|_2$ on an ϵ -net $\mathcal{N}_\epsilon$ is enough to bound it everywhere on the sphere.

Lemma 3 (computing the spectral norm on a net (I))

Let $\mathbf{A}$ be an $m \times n$ matrix, and let $\mathcal{N}_\epsilon$ be an ϵ -net of $\mathcal{S}^{n-1}$ for some $\epsilon \in [0, 1)$. Then

$$\max_{\mathbf{x} \in \mathcal{N}_\epsilon} \|\mathbf{Ax}\|_2 \leq \|\mathbf{A}\|_{\text{op}} \leq (1 - \epsilon)^{-1} \max_{\mathbf{x} \in \mathcal{N}_\epsilon} \|\mathbf{Ax}\|_2$$

Lemma Proof

- The lower bound in the Lemma follows from the **definition of spectral norm**
- To prove the upper bound, we fix $\mathbf{x}^* \in \mathcal{S}^{n-1}$ for which $\|\mathbf{A}\| = \|\mathbf{Ax}^*\|_2$, and choose $\mathbf{y} \in \mathcal{N}_\epsilon$ which approximates $\mathbf{x}^*$ as $\|\mathbf{x}^* - \mathbf{y}\|_2 \leq \epsilon$. By definition of the norm, we have

$$\|\mathbf{Ax}^* - \mathbf{Ay}\|_2 \leq \|\mathbf{A}\| \|\mathbf{x}^* - \mathbf{y}\|_2 \leq \epsilon \|\mathbf{A}\|.$$

By the triangle inequality, it follows that (*recall* $\|\mathbf{A}\| = \|\mathbf{Ax}^*\|_2$)

$$\|\mathbf{Ay}\|_2 \geq \|\mathbf{Ax}^*\|_2 - \|\mathbf{Ax}^* - \mathbf{Ay}\|_2 \geq \|\mathbf{A}\| - \epsilon \|\mathbf{A}\| = (1 - \epsilon) \|\mathbf{A}\|.$$

Taking maximum over all $\mathbf{y} \in \mathcal{N}_\epsilon$ in this inequality, we complete the proof.

□

Computing the spectral/operator norm on a net (II)

Lemma 4 (computing the spectral norm on a net (II))

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric, and let $\mathcal{N}_\epsilon$ be an ϵ -net of the sphere $\mathcal{S}^{n-1}$ with $0 < \epsilon < 1/2$. Then

$$\|\mathbf{A}\|_2 = \sup_{\|\mathbf{u}\|=1} |\mathbf{u}^\top \mathbf{A} \mathbf{u}| \leq \frac{1}{1 - 2\epsilon} \max_{\mathbf{x} \in \mathcal{N}_\epsilon} |\mathbf{x}^\top \mathbf{A} \mathbf{x}|.$$

Proof of Lemma 4 (I)

Goal. We show that for symmetric $\mathbf{A}$ and any ϵ -net $\mathcal{N}_\epsilon \subset \mathcal{S}^{n-1}$ with $\epsilon < \frac{1}{2}$,

$$\|\mathbf{A}\|_2 = \sup_{\|\mathbf{u}\|=1} |\mathbf{u}^\top \mathbf{A} \mathbf{u}| \leq \frac{1}{1 - 2\epsilon} \max_{\mathbf{x} \in \mathcal{N}_\epsilon} |\mathbf{x}^\top \mathbf{A} \mathbf{x}|.$$

- Since $\mathbf{A}$ is symmetric, by the spectral theorem

$$\|\mathbf{A}\|_2 = \sup_{\|\mathbf{u}\|=1} |\langle \mathbf{A} \mathbf{u}, \mathbf{u} \rangle|.$$

- Fix any unit vector $\mathbf{u} \in \mathcal{S}^{n-1}$. By definition of an ϵ -net, there exists $\mathbf{x} \in \mathcal{N}_\epsilon$ with

$$\|\mathbf{u} - \mathbf{x}\|_2 \leq \epsilon.$$

- We expand the difference of quadratic forms:

$$\langle \mathbf{A} \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}(\mathbf{u} - \mathbf{x}), \mathbf{u} \rangle + \langle \mathbf{A} \mathbf{x}, (\mathbf{u} - \mathbf{x}) \rangle.$$

Proof of Lemma 4 (II)

- By Cauchy–Schwarz and $\|\mathbf{u}\| = \|\mathbf{x}\| = 1$, $\|\mathbf{u} - \mathbf{x}\| \leq \epsilon$:

$$|\langle \mathbf{A}(\mathbf{u} - \mathbf{x}), \mathbf{u} \rangle| \leq \|\mathbf{A}\|_2 \epsilon, \quad |\langle \mathbf{A}\mathbf{x}, (\mathbf{u} - \mathbf{x}) \rangle| \leq \|\mathbf{A}\|_2 \epsilon.$$

- Hence

$$|\langle \mathbf{A}\mathbf{u}, \mathbf{u} \rangle| \leq |\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle| + 2\epsilon \|\mathbf{A}\|_2.$$

- Now we take $\sup_{\|\mathbf{u}\|=1}$ and $\max_{\mathbf{x} \in \mathcal{N}_\epsilon}$:

$$\|\mathbf{A}\|_2 \leq \max_{\mathbf{x} \in \mathcal{N}_\epsilon} |\mathbf{x}^\top \mathbf{A} \mathbf{x}| + 2\epsilon \|\mathbf{A}\|_2 \quad \Rightarrow \quad (1 - 2\epsilon) \|\mathbf{A}\|_2 \leq \max_{\mathbf{x} \in \mathcal{N}_\epsilon} |\mathbf{x}^\top \mathbf{A} \mathbf{x}|.$$

- We divide by $(1 - 2\epsilon)$ to conclude the lemma.

$$\|\mathbf{A}\|_2 \leq \frac{1}{1 - 2\epsilon} \max_{\mathbf{x} \in \mathcal{N}_\epsilon} |\mathbf{x}^\top \mathbf{A} \mathbf{x}|.$$

Application to Matrices with Sub-Gaussian Rows

The control of the spectral norm through an ϵ -net allows us to derive **upper and lower bounds on the extreme singular values of a random matrix**, for example when the rows of the matrix are drawn independently from a **sub-Gaussian distribution**:

Theorem 1 (Two-sided bound on sub-Gaussian matrices)

Consider an $m \times n$ matrix $\mathbf{A}$ whose rows $\mathbf{A}_i$ are independent sub-Gaussian isotropic random vectors in $\mathbb{R}^n$. For any $t \geq 0$, we have

$$\sqrt{m} - CK^2(\sqrt{n} + t) \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq \sqrt{m} + CK^2(\sqrt{n} + t) \quad (1)$$

with probability at least $1 - 2\exp(-t^2)$. Here $K = \max_i \|\mathbf{A}_i\|_{\psi_2}$

(a random vector $\mathbf{x}$ is called isotropic if $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = \mathbf{I}_n$)

Recall Approximate Isometry Lemma

Lemma 5 (Approximate Isometry)

Let $\mathbf{A}$ be an $m \times n$ matrix with $\|\mathbf{A}^\top \mathbf{A} - \mathbf{I}\|_2 \leq \delta$.

Then $1 - \delta \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq 1 + \delta$.

How we will use it

We aim to prove (for a suitable δ with high probability)

$$\left\| \frac{1}{m} \mathbf{A}^\top \mathbf{A} - \mathbf{I} \right\|_2 \leq K^2 \delta, \quad \delta = C \left(\sqrt{\frac{n}{m}} + \frac{t}{\sqrt{m}} \right),$$

and then apply Lemma 5 to $\mathbf{A}/\sqrt{m}$ to get

$$\sqrt{m} - CK^2(\sqrt{n} + t) \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq \sqrt{m} + CK^2(\sqrt{n} + t).$$

Theorem 1 – Proof (Step 1: Approximation)

$\frac{1}{4}$ -net and reduction to finitely many directions

By Lemma 2, there exists a $\frac{1}{4}$ -net $\mathcal{N} \subset \mathcal{S}^{n-1}$ with $|\mathcal{N}| \leq 9^n$. Using Lemma 4

$$\|\mathbf{A}\|_2 \leq (1 - \frac{1}{2})^{-1} \max_{\mathbf{x} \in \mathcal{N}} |\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle|,$$

we can upper bound the operator norm of $\frac{1}{m}\mathbf{A}^\top \mathbf{A} - \mathbf{I}$ as follows:

$$\left\| \frac{1}{m}\mathbf{A}^\top \mathbf{A} - \mathbf{I} \right\|_2 \leq 2 \max_{\mathbf{x} \in \mathcal{N}} \left| \left\langle \left(\frac{1}{m}\mathbf{A}^\top \mathbf{A} - \mathbf{I} \right) \mathbf{x}, \mathbf{x} \right\rangle \right| = 2 \max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{m} \|\mathbf{A}\mathbf{x}\|_2^2 - 1 \right|.$$

Thus it suffices to control, uniformly over $\mathbf{x} \in \mathcal{N}$, the scalar deviations $\left| \frac{1}{m} \|\mathbf{A}\mathbf{x}\|_2^2 - 1 \right|$.

Theorem 1 – Proof (Step 2: Concentration – setup)

Fix $\mathbf{x}^* \in \mathcal{S}^{n-1}$ and write the quadratic form as a sum:

$$\|\mathbf{A}\mathbf{x}^*\|_2^2 = \sum_{i=1}^m \langle \mathbf{A}_i, \mathbf{x}^* \rangle^2 =: \sum_{i=1}^m X_i^2,$$

where $\mathbf{A}_i$ is the i -th row of $\mathbf{A}$ and $X_i := \langle \mathbf{A}_i, \mathbf{x}^* \rangle$.

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where $\mathbf{A}_i$ is the i -th row of $\mathbf{A}$ and $X_i := \langle \mathbf{A}_i, \mathbf{x}^* \rangle$.

Assumptions transferred to X_i

Each row $\mathbf{A}_i$ is independent, isotropic, sub-Gaussian with $\|\mathbf{A}_i\|_{\psi_2} \leq K$. Hence

X_i independent, isotropic ($\mathbb{E} X_i^2 = 1$), sub-Gaussian with $\|X_i\|_{\psi_2} \leq K$.

Facts: $Z_i := X_i^2 - 1$ are **sub-exponential** with $\|Z_i\|_{\psi_1} := \inf\{s > 0 \mid \mathbb{E} e^{Z_i/s} \leq 2\} \leq CK^2$.

$\frac{1}{m} \|\mathbf{A}\mathbf{x}^*\|_2^2 - 1 = \frac{1}{m} \sum_{i=1}^m Z_i$. We apply Bernstein's inequality to the average of i.i.d. sub-exponential r.v.

Theorem 1 – Proof (Step 2: Concentration – Bernstein)

Bernstein for sub-exponential averages

If Z_i are i.i.d. with $\|Z_i\|_{\psi_1} \leq \nu$, then for all $u > 0$,

$$\Pr\left(\left|\frac{1}{m} \sum_{i=1}^m Z_i\right| \geq u\right) \leq 2 \exp\left[-c m \min\left(\frac{u^2}{\nu^2}, \frac{u}{\nu}\right)\right].$$

Here $\nu = CK^2$.

Set $u = \varepsilon/2$. Then for each fixed $\mathbf{x}^* \in \mathcal{S}^{n-1}$,

$$\Pr\left(\left|\frac{1}{m} \|\mathbf{A}\mathbf{x}^*\|_2^2 - 1\right| \geq \frac{\varepsilon}{2}\right) \leq 2 \exp\left[-c_1 m \min\left(\frac{\varepsilon^2}{K^4}, \frac{\varepsilon}{K^2}\right)\right].$$

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Reparametrization

Choose ϵ such that

$$\frac{\varepsilon}{K^2} = \delta, \quad \delta = C\left(\sqrt{\frac{n}{m}} + \frac{t}{\sqrt{m}}\right).$$

Then $\min\left(\frac{\varepsilon^2}{K^4}, \frac{\varepsilon}{K^2}\right) = \min(\delta^2, \delta) =: \delta_2$, and

$$\Pr\left(\left|\frac{1}{m}\|\mathbf{A}\mathbf{x}^*\|_2^2 - 1\right| \geq \frac{\varepsilon}{2}\right) \leq 2 \exp(-c_1 \delta_2 m).$$

Theorem 1 – Proof (Step 2: Concentration – Bernstein)

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$$\Pr\left(\left|\frac{1}{m}\|\mathbf{A}\mathbf{x}^*\|_2^2 - 1\right| \geq \frac{\epsilon}{2}\right) \leq 2 \exp(-c_1 \delta_2 m).$$

Finally, with $\delta = C(\sqrt{n/m} + t/\sqrt{m})$, $\delta_2 m \geq C^2(n + t^2)$ (up to constants; using $a^2 + b^2 \leq (a + b)^2$), we get

$$\Pr\left(\left|\frac{1}{m}\|\mathbf{A}\mathbf{x}^*\|_2^2 - 1\right| \geq \frac{\epsilon}{2}\right) \leq 2 \exp[-c_1 C^2(n + t^2)].$$

Theorem 1 – Proof (Step 3: Union bound over the net)

By the union bound,

$$\Pr\left(\max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{m} \|\mathbf{A}\mathbf{x}\|_2^2 - 1 \right| \geq \frac{\varepsilon}{2} \right) \leq |\mathcal{N}| \cdot 2 \exp[-c_1 C^2(n + t^2)] \leq 2 \cdot 9^n \exp[-c_1 C^2(n + t^2)].$$

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Choosing C (absorbing 9^n)

Pick C large enough so that $2 \cdot 9^n \exp[-c_1 C^2 (n + t^2)] \leq \exp(-t^2)$. Then

$$\Pr\left(\max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{m} \|\mathbf{A}\mathbf{x}\|_2^2 - 1 \right| \geq \frac{\varepsilon}{2} \right) \leq \exp(-t^2).$$

By Step 1,

$$\left\| \frac{1}{m} \mathbf{A}^\top \mathbf{A} - \mathbf{I} \right\|_2 \leq 2 \max_{\mathbf{x} \in \mathcal{N}} \left| \frac{1}{m} \|\mathbf{A}\mathbf{x}\|_2^2 - 1 \right| \leq \varepsilon$$

with probability at least $1 - e^{-t^2}$.

Wrap-up: From quadratic concentration to singular values

We have shown that, with probability at least $1 - e^{-t^2}$,

$$\left\| \frac{1}{m} \mathbf{A}^\top \mathbf{A} - \mathbf{I} \right\|_2 \leq \varepsilon, \quad \varepsilon = K^2 \max(\delta, \delta^2), \quad \delta = C \left(\sqrt{\frac{n}{m}} + \frac{t}{\sqrt{m}} \right).$$

Wrap-up: From quadratic concentration to singular values

We have shown that, with probability at least $1 - e^{-t^2}$,

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Apply Lemma 5 to $\mathbf{A}/\sqrt{m}$:

$$1 - \varepsilon \leq s_{\min}\left(\frac{\mathbf{A}}{\sqrt{m}}\right) \leq s_{\max}\left(\frac{\mathbf{A}}{\sqrt{m}}\right) \leq 1 + \varepsilon.$$

Multiply by $\sqrt{m}$:

$$\sqrt{m} - \varepsilon\sqrt{m} \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq \sqrt{m} + \varepsilon\sqrt{m}.$$

Since $\varepsilon\sqrt{m} = K^2 \max\left(C(\sqrt{n} + t), \frac{C^2}{\sqrt{m}}(n + t^2)\right) \lesssim K^2(\sqrt{n} + t)$, we obtain

$$\sqrt{m} - CK^2(\sqrt{n} + t) \leq s_{\min}(\mathbf{A}) \leq s_{\max}(\mathbf{A}) \leq \sqrt{m} + CK^2(\sqrt{n} + t)$$

with probability at least $1 - e^{-t^2}$.